An Identified-Model, Estimator-Consistency and Requirements Re-Baseline of a One-Axis CubeSat ADCS Testbed

A re-analysis of the SPACE 584 Fall 2025 final report

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Abstract

The SPACE 584 FA2025 class built a 1U-class CubeSat with a reaction wheel, two magnetorquers, a star tracker, a thermal-array sun sensor and a rotating test platform, and reported that most of its top-level goals were missed. This paper re-examines that work using recomputed numbers, a first-principles model identified from the report's own data (digitised from its raster figures), a redesigned estimator and controller evaluated by deterministic Monte Carlo, and a re-baselined requirement set. The audit found 29 items (4 critical, 15 high). Among them are an inertia error of a factor 2.6 that propagates into the wheel-resolution requirement, which is itself off by a factor of ten in units; a detumble requirement that bearing friction alone satisfies (28 s); a gyro that was saturated at its 500 dps range in the headline test; and a Helmholtz/torquer functional test that cannot distinguish magnetic action from friction. We identify the testbed friction (Coulomb torque 4.1×10^{-4} N m, validated to 10% on independent data) and show that the wheel channel was acceleration-limited rather than throttle-proportional. A gyro/sun/star EKF with a bias state and delay handling is NEES/NIS-consistent (mean NIS 1.004), whereas the report's filter is over-confident by about two orders of magnitude under realistic sensor noise. Replacing the state machine with a rate/attitude controller raises the simulated mission success within 60 s from 1% [0, 3] to 100% [98, 100], and a time-optimal slew is 13% [11, 14] faster than the braking-profile PID for 90°. A strict novelty screen against the literature reclassifies most candidate improvements as replications, and two of them (star-field design for uniqueness, coordinated wheel/torquer optimisation) did not hold and are dropped. The contribution that can be supported is a documented identified-model and estimator-consistency study of a student testbed, with a complete v2 requirement set, budgets and test campaign. All numbers regenerate from `run_all.py`.

Nomenclature

Dynamics and actuators		*Estimation and sensing*	
θ, ω	body attitude [rad] and rate [rad/s]	b (EKF)	gyro bias state [rad/s]
I_z, I_{cs}	CubeSat inertia about the spin axis [kg m^2]	P, Q, R	state, process and measurement covariances
I_p, I_t	platform and total (loaded) inertia [kg m^2]	ν, S	innovation and its covariance
I_w, Ω	wheel inertia and speed	τ	measurement latency [s]
q	wheel momentum in body-rate units [rad/s]	NEES, NIS	normalised estimation error / innovation squared
T_c, b (friction)	Coulomb torque [N m], viscous coefficient	IFOV	instantaneous field of view [μrad]
τ_e, r_{max}	wheel lag [s], acceleration limit [rad/s^2]	t_e	exposure time [s]
m, B, μ_{eff}	dipole [A m^2], field [μT], effective permeability	s	smear [px]

1 Introduction

Attitude determination and control is one of the subsystems that most often limits a CubeSat mission, and a ground testbed is the usual way for a student team to exercise it before flight. In the Fall 2025 SPACE 584

course the class designed, built and tested a one-axis ADCS housed in a 1U-class structure, together with the star field, Helmholtz coil and rotating platform needed to test it. The final report [1] documents that work honestly, and also reports that most of the top-level goals were missed. The author contributed the CAD, the sensor testing and the state controller to that team report.

The report leaves open *why* the goals were missed. Several of its numbers can be recomputed from its own equations, and its figures contain the only measurements of the testbed's dynamics, but neither had been checked. The inputs to this study are the 53-page final report of SPACE 584 (FA2025) [1], its public code repository (branch `integrated_adcs`) [2], and the author's round-1 notes. The repository was cloned into an isolated folder and read as data only; no code from it was executed. Where the report's figures contain the only available measurements (Figs. 37 and 38), they were digitised from the raster images embedded in the PDF using tick-detected axis calibration (resolution ≈0.05 rad/s, 0.03 s).

The goal of this paper is to determine which of the report's conclusions survive a full recomputation, and to give a baseline that a v2 team could build and test. To do this, the study (1) audits the report (§3); (2) identifies a dynamic model of the platform, CubeSat and wheel from the report's data (§4); (3) redesigns the controller and estimator and tests them by Monte Carlo (§5–6); (4) re-examines star identification on the *real* star-field map (§7); (5) redesigns the magnetic subsystem and the test coil (§8); (6) analyses imaging and computes the spin-rate thresholds the report left open (§9); (7) screens candidate novelties against the literature (§10); and (8) gives a v2 requirement set, budgets and test campaign (§11).

The claims of this paper are stated strictly, as follows:

- *Replicated*: time-optimal slews, gyro-bias estimation from optical fixes, delayed-measurement updates, wheel/magnetorquer momentum management, star-ID voting and gyro-aided stacking. All of these exist in the literature (§10); here they are quantified for this testbed.
- *Improved over the report*: the arithmetic and unit corrections, the identified model, a consistent estimator, the controller, the thresholds, the requirement set, the drive electronics and the coil geometry.
- *Novel (supportable only)*: a documented *identified-model plus estimator-consistency study of a student testbed*, built from the testbed's own digitised data, with its provenance and uncertainty. We claim no new algorithm.
- *Dropped*: star-field design for uniqueness, the coordinated LQ wheel/torquer scheme as an improvement, and the faster-star-ID claim.

The status of every input is tracked as VERIFIED (read from a datasheet or paper), DERIVED (computed from verified values), or UNVERIFIED/assumed (flagged in the text and in `research/params.json`).

The rest of the paper is structured as follows. Section 2 describes the approach to the data, its provenance and the lumped CAD model. Section 3 presents the audit of the report, and Sections 4–9 develop the identified model, the controller, the estimator, star identification, the magnetic subsystem and imaging. Section 10 screens the candidate improvements against the literature, and Section 11 gives the v2 baseline and test campaign. The paper then discusses the lessons learned and the sources of error and model limitations (Section 12), and concludes with an assessment of what the testbed can and cannot demonstrate (Section 13).

2 Approach: Data, Provenance and the Lumped CAD Model

Every number in this paper is either read from a source, derived from read values, or assumed, and the analysis is built so that each can be regenerated from the report and the data. Two inputs underpin most of what follows: the digitised measurements and a lumped CAD model for the inertia.

Digitisation. Fig. 38 (rate, throttle, attitude) and Fig. 37 (ST/SS angle) of the report are embedded as rasters (2264×1590 and 2200×535 px). Axis ticks are detected from the frame, a linear map is fitted (residual <2 px), and the blue and orange series are tracked by colour. The gyro curve resolves ≈0.05 rad/s. The Hall curve is piecewise linear between sparse pulses (one per revolution), so it is used only as an independent check.

Lumped CAD model. The layer stack (PETG base 4 mm, three 1.6 mm Al plates, four M2 brass standoffs, 100×150 mm) is built in `cadquery`. The solid inertia comes from the CAD solids, and the component masses are catalogue values (flagged as assumed), with the battery mass taken as the remainder to 960 g. That remainder is ≈235 g, i.e. one 4S pack (Fig. 1). A Monte Carlo over ±15% in component mass and position gives the I_z band quoted in §3 and a CoM offset of 2.9 mm.

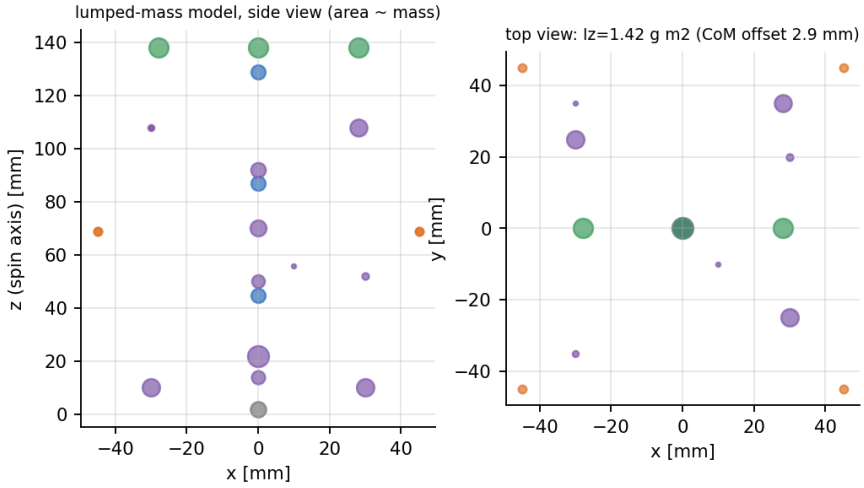


Figure 1: Lumped-mass model used for the inertia estimate (marker area $\propto$ mass). The battery is the remainder to the measured 960 g.

3 Audit of the Report

The audit is generated by `audit/audit_checks.py` and lists, for each item, the claim, the paper value, the recomputed value, the effect and a fix. Table 1 gives the ranking, and the most consequential findings are developed below. Figure 2 summarises the count by severity together with the v2 budgets used later.

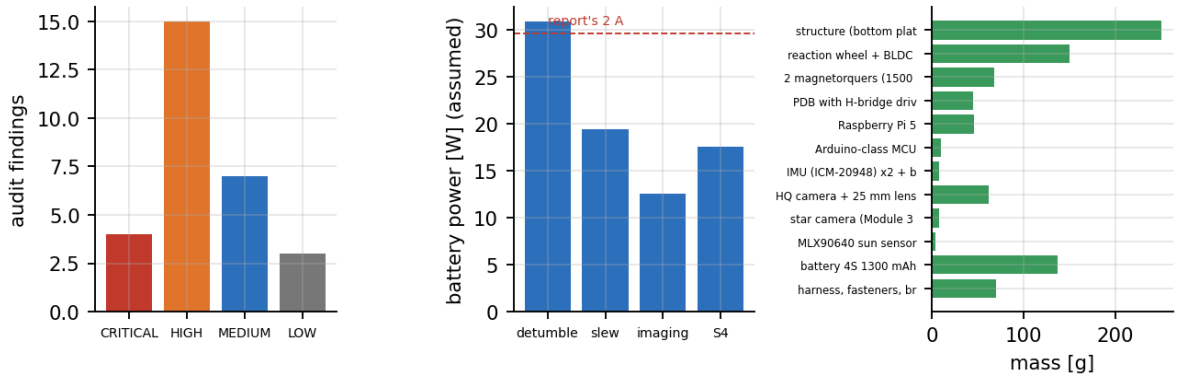


Figure 2: Audit findings and v2 budgets. Left: audit findings by severity. Middle: v2 battery power per mode (assumed loads) against the 2 A the report sized for. Right: v2 mass budget.

3.1 Headline Findings

A01 Inertia (critical). The report sizes the wheel with $I_{sat} = 0.5 \cdot 1.5 \cdot 0.05^2 = 0.00375\ \mathrm{kg\,m^2}$. The product is in fact 0.001875, a factor-2 arithmetic slip. The model also uses 1.5 kg for a 0.96 kg spacecraft, and a disk for what is a square prism. A parametric CAD/lumped-mass model (`cad_inertia.py`, 960 g total, battery mass as the remainder) gives $I_z = 1.42 \times 10^{-3}\ \mathrm{kg\,m^2}$ (5–95%: 1.38–1.46), against closed-form bounds of 1.2–1.6 (disk, 10×10 prism). The wheel therefore needs 133 rad/s (1270 RPM), not 312 rad/s, to absorb 1 rev/s, and the inertia ratio is ≈19–21, not 49.6.

A02 Resolution requirement (critical). A rate of 0.001 rad/s is 0.00955 RPM, not 0.0942 RPM, so the requirement is ten times too large. With the corrected ratio the wheel step must be ≈ 0.2 RPM. The 10-bit ESC step of 13.67 RPM is ≈ 70× too coarse, and the proposed 12-bit step (3.42 RPM) is still ≈ 17× too coarse, which contradicts the report's conclusion. About 16 bits (or a closed speed loop) removes the issue. There is, however, a nuance found in §5: a fixed-step state machine cannot settle below one bin, but closed-loop rate feedback dithers across bins and still reaches a few mrad/s rms on the same coarse wheel. Resolution is therefore as much a control-logic problem as a hardware one.

A03 Requirement A is satisfied by friction (critical). The platform stops a mounted CubeSat from

1 rev/s in 37.8 s with no ADCS (Table 12 of the report), and the identified friction brings it below 10°/s in 28 s. As a result, the requirement "1 rev/s→10°/s in 60 s" cannot discriminate a working ADCS from a dead one.

A04 Gyro saturation (critical). In Fig. 38 the gyro trace is flat at −8.70 rad/s for $t < 11$ s, while the Hall sensor reaches −10.6 rad/s. Since 500 dps equals 8.727 rad/s, and the team's sketch sets `ICM20948_GYRO_RANGE_500_DPS` [2, 3], the gyro was saturated. The test started at 1.7 rev/s, not 1 rev/s, and the first 10 s of rate feedback was clipped. The text's "up to 2.1 rad/s in the opposite direction" also disagrees with the plotted peaks of 2.81 (gyro) and 2.69 (Hall).

A07/A08 Smear budget. The rate requirement uses 0.5×0.0002 rad, but 0.0002 rad is the 1 mm resolution (3.2 px). The HQ-camera IFOV with the 25 mm lens is 62 μrad (1.55 μm pixels [4]), so 0.5 px is 31 μrad, i.e. 3.1×10^{-4} rad/s at the same 0.1 s exposure and 1.9×10^{-3} rad/s at 1/60 s. The test images taken at 0.5–2.5 rad/s carried 134–672 px of smear at 60 fps, about 269× the budget. The failure of the 1 mm objective is therefore explained by motion, and the statement that the 'focus shifted when in motion' is not supported.

A09–A11 Magnetorquers. A relay is an on/off device with no bipolar drive or flyback path. The 493-turn rod is 3.4 Ω, draws 1.47 A at 5 V and stores 6.0 mH. The report's thermal threshold (0.01 K/s at 9 V) is off by a factor of ∼200: the recomputed heating is 0.70 K/s at 5 V and 2.3 K/s at 9 V. Because $m = \mu_{eff} NIA$ with $I = V/R \propto 1/N$, the dipole is independent of the number of turns at fixed voltage. The rewinding to ≈3400 turns on a shorter rod therefore lowered the modelled dipole from 1.14 to 0.68 A m^2 and raised the mass from 22 to 61 g.

A13 De-saturation requirement (high). A wheel speed of 200 rev/s (12 000 RPM) exceeds the wheel's own 7000 RPM, and removing the stated 0.090 N ms in 120 s needs 0.75 mN m, i.e. 8.4 A m^2 at 90 μT, against the 0.95–1.6 delivered. The target is physically infeasible for this hardware.

A14–A16 Helmholtz coil. A square pair needs a separation of 0.546 of the side. The built 14 in separation on 12 in coils gives a 26% peak-to-peak variation over the body (10% at the right spacing). The stated wire gives 83 Ω and 0.06 A at 5 V, but 0.97 A was measured. Table 10's field falls by 63% (factor 2.7) between sensors 39 mm apart, the resistance rises by 19% (≈39 K), and the cross-axis coupling is 15%. The functional test ("stopped in 50 s") is not evidence of magnetic detumbling: friction stops the loaded platform in 38 s, and the available magnetic torque is only 0.21 of the friction torque.

A18 Power. A current of 2 A for 3 min is 100 mAh, so "over 2200 mAh" is 22× what the arithmetic requires. Three 4S 2200 mAh packs weigh 717 g (verified pack masses 232–245 g [5]), and the lumped mass budget leaves room for only one. The 30 s run time is therefore not a capacity problem; a brown-out is the likely, but undiagnosed, cause.

A20–A22 Sensors and estimation. The digitised Fig. 37 shows one star-tracker and one sun-sensor sample every 6.4 s (0.16 Hz, not the stated 0.5 Hz), 15 distinct sun-sensor levels in 160 s, and an RMS ST–SS discrepancy of 5.0° (max 9.5°). The EKF's 5 nT magnetometer noise corresponds to 0.010°, whereas one LSB of the ICM-20948 magnetometer (0.15 μT [3]) is 0.30°. Details are given in §6 and §7.

Table 1: Audit findings ranked by severity (full text, paper vs recomputed values and fixes in `audit/audit.md`).

ID	Severity	Area	Finding (short)
A01	Critical	Reaction wheel sizing	The product 0.5*1.5*0.05^2 is 0.001875, not 0.00375 (a factor-2 arithmetic slip).
A02	Critical	Reaction wheel resolution	Unit conversion error: 0.001 rad/s is 0.00955 RPM, ten times smaller than the 0.0942 RPM used.
A03	Critical	Detumble requirements / test validity	The testbed's own bearing friction (identified from Tables 11-12, torque ~4e-4 N m) brings a mounted CubeSat from 1 rev/s to 10 deg/s in about 28 s and to rest in 38 s with the ADCS off.
A04	Critical	Gyroscope / attitude control test	In Fig. 38 the gyro trace is flat at -8.70 rad/s for t<11 s while the Hall sensor continues to rise to -10.6 rad/s.
A05	High	Attitude control test	The plotted peak is 2.8 rad/s (gyro) and 2.7 rad/s (Hall) at t = 45 s, 30-35 % above the quoted value..

ID	Severity	Area	Finding (short)
A06	High	Attitude control test	Momentum bookkeeping on the gyro trace (friction removed) shows the wheel accelerating at a nearly constant ~0.3 rad/s^2 (body-equivalent) regardless of the throttle step (sensitivity 0.04 rad/s^2 per 100 % throttle, bootstrap 95 .
A07	High	Camera / smear budget	0.0002 rad is the 1 mm resolution requirement (3.2 px), not the pixel IFOV.
A08	High	Camera / imaging test	At 0.5 rad/s and a 60 fps exposure the smear is 134 px (about 40 mm on the target) against a 0.5 px budget: 270x over.
A09	High	Magnetorquer drive electronics	A relay is an on/off device: it cannot realise a proportional dipole from 'varying PWM', has no bipolar drive (B-dot needs both polarities), contact bounce/wear at PWM rates, and without a flyback path the ~6 mH coil (0.01 J scale.
A10	High	Magnetorquer thermal sizing	With R = 3.4 Ω (0.34 Ω/m x ~10 m), P = V^2/R and a heat capacity of ~11 J/K (rod 17.8 g steel + wire 5.6 g Cu), heating is 0.7 K/s at 5 V and 2.3 K/s at 9 V, 70-230x the quoted values.
A11	High	Magnetorquer sizing	At fixed voltage the dipole m = mu_eff*N*I*A is independent of N (I = V/R and R grows with N), exactly as the report itself shows for the Helmholtz coil.
A13	High	Momentum dump requirement	The wheel as built cannot reach 12000 RPM (the report's own limit is 7000 RPM), and dumping the stated momentum in 120 s needs 0.75 mN m, i.e. 8.4 A m^2 at the 90 uT test field: 5-9x more than the torquers deliver.
A14	High	Helmholtz coil design	Square Helmholtz pairs need a separation of 0.5445-0.546 of the side (here 6.6 in); 14 in is 2.1x too wide and gives ~26 % peak-to-peak variation over the 10 x 10 x 15 cm body (10 % at the right spacing).
A15	High	Helmholtz calibration (Table 10)	The field falls by 2.7x (63 %) between sensors 39 mm apart, as for a near-wire field (~1/r^1.8), not as in the uniform centre of a 305 mm coil; so the sensors were not at the centre, or the coil is not uniform.
A16	High	Helmholtz functional test	Bearing friction alone stops the loaded platform from 1 rev/s in 38 s (Table 12).
A18	High	Power / battery sizing	2 A for 3 min is 100 mAh; the 2200 mAh figure is 22x the report's own arithmetic.
A20	High	Sun sensor / star tracker accuracy claims	Digitised markers in Fig. 37 show one SS sample every 6.4 s and one ST sample every 6.4 s (0.16 Hz), not 0.5 Hz.
A21	High	EKF assumptions and validation	5 nT of magnetometer noise corresponds to 0.010 deg in heading but the ICM-20948 magnetometer resolution is 0.15 uT (0.30 deg): the assumed noise is 30x finer than one LSB.
A22	High	Star identification / artificial field	On the real 56-star map, 99 % of pair distances have a coincident partner within the 0.004 rad tolerance and 92 % of stars share their first-neighbour distance with another star, so the unique-match claim rests entirely on later c.
A12	Medium	Magnetorquer test	0.95/1.6 = 59 % of the target, i.e. 41 % low.
A17	Medium	Rotating platform data (Tables 11-12)	Spin-down times are not monotone in speed in Table 11 (0.84 -> 4.9 s, 1.04 -> 7 s, 1.39 -> 7.3 s) and the best-fit Coulomb model leaves ~20 % log residuals.
A19	Medium	Camera resolution	The static claim is consistent: 0.7 mm = 2.3 px of the 0.31 mm GSD (Nyquist limit 0.62 mm), FOV 14.3 x 10.8 deg agrees.
A23	Medium	Requirements table	(i) 200 rev/s (12000 RPM) exceeds the 7000 RPM / 117 rev/s wheel by 1.7x; (ii) Table 1 'slow to 30 deg/s' and Table 6 'less than 0.5 deg/s after 2 minutes using wheels and rods' are mutually inconsistent in time bases (60 s vs 2 m.

ID	Severity	Area	Finding (short)
A24	Medium	State machine / thresholds	The thresholds follow from exposure, IFOV and update period (a6): sun-sensor acquisition <= 0.38 rad/s at 2 Hz updates but only 0.04 rad/s at the measured 6.6 s period; star tracker <= 3.9 mrad/s (1 px, 0.1 s) to 0.058 rad/s (3 px.
A25	Medium	Mass / volume	150 mm is 1.3-1.5U, not 2U (2U = 227 mm); the 1 kg limit is a course constraint, not the CubeSat spec (Rev 14.1: 2 kg for 1U); the two mass figures differ by 10 g without a stated measurement uncertainty; the volume requirement wa.
A26	Medium	Hall sensor / Fig. 38 agreement	With one Hall pulse per revolution the sensor gives ~1 sample per 2.5-12 s once the platform slows; the orange curve is a straight-line interpolation (vertices visible) and differs from the gyro by up to 1 rad/s between vertices.
A27	Low	State numbering / telemetry	The two enumerations disagree (S3 is science imaging in one and desaturation in the other); Table 13 lists 33 doubles (indices 0-32) but states 35 doubles/280 bytes..
A28	Low	Cost / schedule	$6000 - $3000 leaves $3000, not a $4000 margin; the requirement as written is ambiguous..
A29	Low	Wheel speed arithmetic	311.6 rad/s is 2976 RPM; 'KeV' should be KV.

4 Identified Platform and Wheel Model

4.1 Bearing Friction

For a Coulomb friction torque T_c and a viscous coefficient b, the time to stop from ω_0 is

$$t_s = \frac{I}{b} \ln\left(1 + \frac{b\,\omega_0}{T_c}\right), \tag{1}$$

or $I\omega_0/T_c$ for $b = 0$, where I is the rotating inertia. The report's Tables 11–12 give nine stop times (six unweighted, three with the CubeSat). We fit T_c, b and the platform inertia I_p (with I_{cs} from CAD and its $\pm$ range of 1.2–1.6 g m^2 propagated) by least squares on the log-times. The Coulomb-only model is preferred by AICc, with $T_c = 4.1 \times 10^{-4}$ N m (95% bootstrap 2.7–6.9), $I_p = 0.45 \times 10^{-3}$ kg m^2 and a loaded inertia of 1.87×10^{-3}. The residual is 24% in log-time: the data are noisy (non-monotone in Table 11) and a viscous term cannot be separated. As an independent validation, the model applied to the pre-wheel segment of Fig. 38 (Hall rate $-10 \to -7.8$ rad/s) predicts a deceleration of 0.221 against 0.200 rad/s^2 measured (Fig. 3).

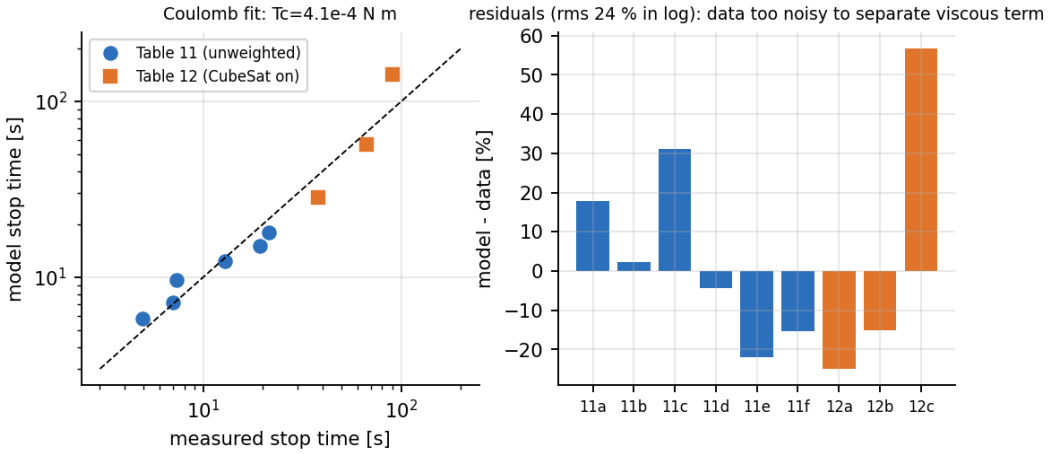


Figure 3: Platform spin-down model versus the report's stop-time data. Left: model versus measured stop times. Right: relative residuals for Table 11 (blue) and Table 12 (orange).

4.2 Wheel and ESC Channel

Let $q = -I_w\Omega/I_t$ be the wheel momentum in body-rate-equivalent units. Momentum conservation then gives

$$\dot{\omega} = \dot{q} - f\,\mathrm{sgn}(\omega), \tag{2}$$

where $f = T_c/I_t$. The *model-free* momentum trace $q(t) = \omega(t) - \omega(t_0) + \int f\,\mathrm{sgn}\,\omega\,dt$ shows (Fig. 4) that the wheel gains momentum at an almost constant 0.31 rad/s^2 over the whole staircase, and still holds 8.4 rad/s-equivalent at 60 s, although the report states it was 'off'. A parametric fit with a throttle-sensitivity term s gives $s = 0.04$ rad/s^2 per unit throttle (95% block-bootstrap [-0.05, 0.14]; ΔAICc=5.4), which is consistent with throttle having no effect, with a residual rms of 0.25 rad/s. The first '0%' step had already consumed most of the momentum authority ($\tau = 0.42$ s transient). The equivalent wheel torque is 5.8×10^{-4} N m and the peak held momentum is 13.2 rad/s-equivalent ($\approx$ 3115 RPM if I_w and I_t are right). This directly shows that the wheel was acceleration-limited and that the throttle-to-speed map was never exercised.

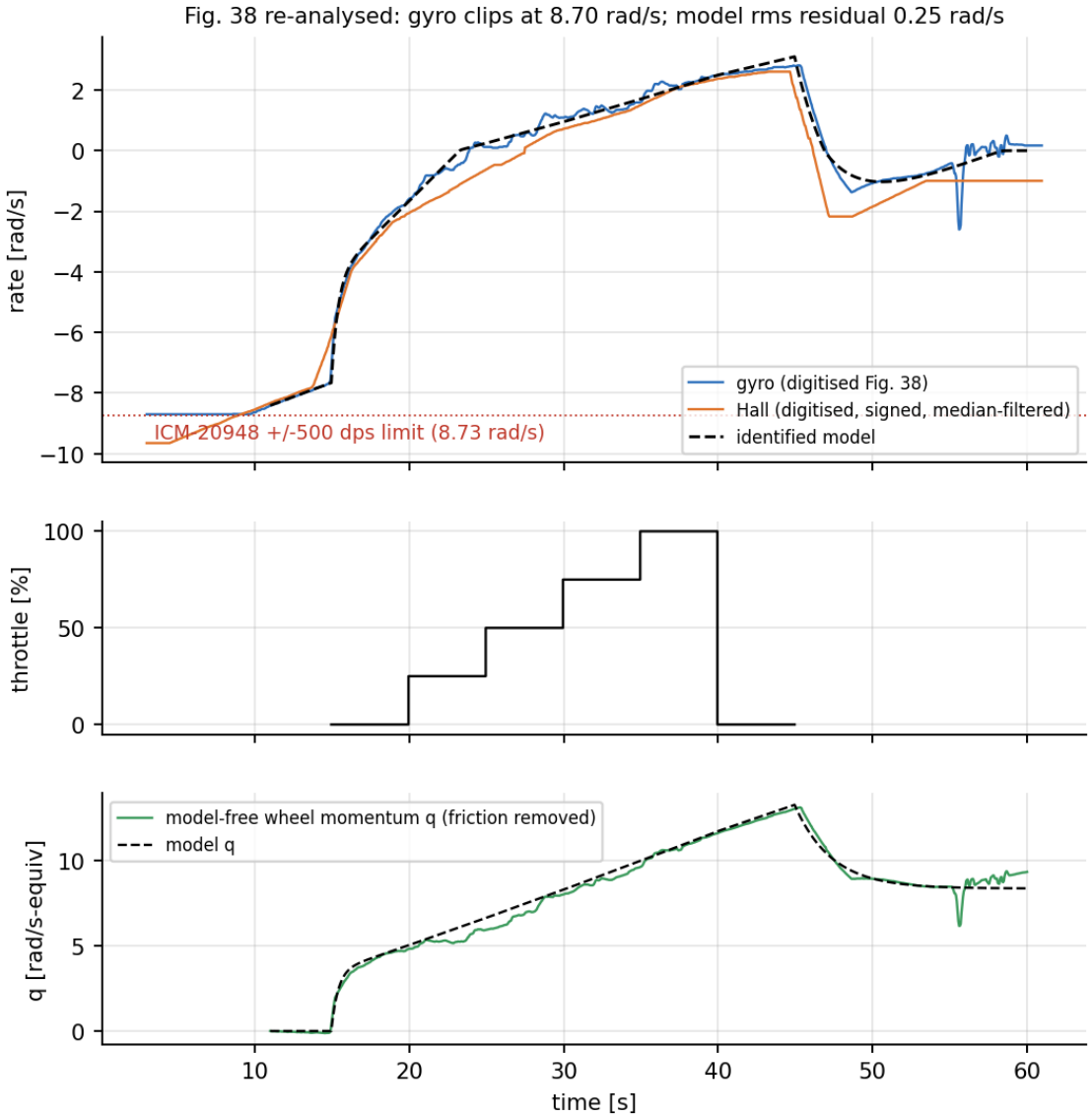


Figure 4: Re-analysis of the report's Fig. 38. Top: digitised gyro (clipped at −500 dps), Hall and identified model. Middle: throttle staircase. Bottom: wheel momentum, model-free and model. The momentum rises at a near-constant rate regardless of throttle.

5 Controller Redesign

5.1 Plant, Sensors and Estimator

The simulator (`simlib.py`) integrates

$$\dot{\theta} = \omega, \qquad \dot{\omega} = \dot{q} - f\tanh(\omega/0.03) + d, \tag{3}$$

where d is a disturbance torque per unit inertia, with the wheel modelled as $\dot{q} = \mathrm{clip}((q_{cmd} - q)/\tau_e, \pm r_{max})$, $|q| \le q_{max}$, and a command resolution Δq. For the v1 hardware we use $\Delta q = 0.058$ rad/s-eq (13.67 RPM), $r_{max} = 0.31$ rad/s^2 (identified), a gyro range of ± 500 dps, and optics every 6.6 s with a 4 s star-tracker delay and a 6% misidentification rate. For v2 we use $\Delta q = 4 \times 10^{-4}$, $r_{max} = 0.53$ (1 mN m), a gyro range of ± 1000 dps (a verified option [3]), and 2 Hz optics with 0.3 s latency. The gyro noise is 0.015 dps/$\sqrt{\text{Hz}}$ over 51 Hz [3], the initial zero-rate offset is $\sigma = 3$ dps (datasheet ± 5 dps), and a bias drift of 5×10^{-5} rad/s/$\sqrt{\text{s}}$ is assumed. Optical fixes are gated by the fields of view (sun pad at $-12°$, $\pm 27.5°$; star field $-66°$ to $8°$ plus the crop) and by blur (the star tracker is unusable above 0.02 rad/s), and their noise includes smear. All controllers share the same EKF (§6) and a search mode (a sweep at 0.3 rad/s when the attitude is unknown), which the report lacked.

5.2 Controllers

Four controllers are compared.

Report state machine (ported): S0 is gyro-only rate stepping by one wheel bin per 0.5 s tick, S1 is coarse pointing at < 0.3 rad/s, S2 is fine pointing and S3 is imaging, with transition thresholds of 0.3 and 0.05 rad/s as in the report's Fig. 8.

Cascade PID: the attitude loop commands $\omega_{ref} = -\mathrm{sgn}(e)\min(K_\theta|e|, \sqrt{2 \cdot 0.6\, a_{max}|e|})$ (a braking-capable profile), followed by a rate PI with back-calculation.

LQR/LQG: $u = -K[\theta_e, \omega, \int \theta_e]$, obtained from the algebraic Riccati equation for the triple integrator with a torque-limit weight $R = 1/a_{max}^2$ and the error clipped for saturation.

Time-optimal: bang-bang control on the switching curve $s = e + \omega|\omega|/(2a) + \omega(\tau_e + 0.04)$ with $a = 0.9\, a_{max}$, handing the hold phase to the LQR.

The gains were tuned on a 30° slew and hold, using a grid over $K_\omega \in \{2, 5, 10\}$, $K_\theta \in \{0.5, 1, 2\}$ for the PID and over weight/effort ratios for the LQR. The selected values are $K_\omega = 5$ and $K_\theta = 2.0$, and the LQR gain is $K = [3.47, 3.08, 0.32]$.

Stability margins. For the loop with a wheel lag of 0.05 s and a 50 ms delay, the PID rate loop has a phase margin of 52° and a gain margin of 13 dB, the attitude loop 63°/11 dB, and the LQR 52°/49 dB. Anti-windup is by back-calculation (PID) and by clipping of the error and integrator (LQR), and the momentum limit is enforced in the wheel model.

5.3 Results

Table 2 gives the full-mission Monte Carlo ($N = 300$ initial conditions per cell, seed-fixed). With the as-built wheel, the report's state machine succeeds in 1% [0, 3] of trials (rms rate 13.6 mrad/s against the 6.2 mrad/s imaging limit). One wheel bin (0.058 rad/s) exceeds the S1 cut-off and the imaging limit is 9× below one step, so the logic cannot converge. A properly retuned PID on the *same* hardware reaches 26% [22, 32], and v2 hardware reaches 100% [98, 100] (PID), 97% [95, 99] (LQR) and 99% [97, 99] (time-optimal), with median ready times of 21.4 [20.4, 22.5], 24.7 [22.9, 26.4] and 21.2 [19.7, 22.3] s. With the attitude known (as in the report's own Simulink test) and friction removed, the report's logic with 13.7 RPM bins reaches imaging in 1% [0, 4] of trials, a v1-resolution PID in 95% [90, 97] with an rms rate of 3.6 mrad/s, and v2 hardware in 100% [98, 100] (rms 4.6 mrad/s). In the v1 case the feedback dithers across the wheel bins, so quantisation alone is not fatal to a closed-loop controller; it is fatal to a fixed-step state machine. Figures 5 and 6 show example trajectories and the success statistics.

Detumbling from 1–1.7 rev/s is torque-limited and nearly identical for every controller (≈15 s). The differences appear in rest-to-rest slews (Table 3): the time-optimal controller is 13% [11, 14] faster than the PID at 90° and 10% [9, 11] faster at 150°, and 11% [10, 11] / 12% [12, 12] faster than the saturated LQR, which overshoots by 3.5° at 90°. Against a 3-s, 0.4 rad/s^2 step disturbance on an air bearing, the peak deviation is 1.7° (PID), 6.2° (LQR) and 3.0° (time-optimal).

Table 2: Full-mission Monte Carlo (N=300, random initial spin 0.5–1.7 rev/s and angle, EKF + delayed optics + search mode, test-bed friction on). 'Ready' = within ±3.5° and below 6.2 mrad/s for 1 s. Median [95% bootstrap CI]; success is Wilson 95%.

Hardware	Controller	P(ready≤60 s)	t_{ready} median [s]	rms ω last 10 s [mrad/s]
v1	report SM	1% [0, 3]	22.8 [14.3, 27.0]	13.6
v1	PID (v1 gains)	26% [22, 32]	62.9 [60.6, 64.6]	28.4
v2	cascade PID	100% [98, 100]	21.4 [20.4, 22.5]	0.7
v2	LQR/LQG	97% [95, 99]	24.7 [22.9, 26.4]	0.6
v2	time-optimal	99% [97, 99]	21.2 [19.7, 22.3]	1.5

Table 3: Rest-to-rest slew settle time (air-bearing plant, attitude known to 0.3°, N=200). Median [95% CI] and overshoot; $t_{min} = 2\sqrt{\theta/a}$ is the bang-bang bound for $a = 0.9\,a_{max}$.

Slew	PID [s]	LQR [s]	time-optimal [s]	TO gain vs PID	t_{min} [s]
30°	4.18 [4.12, 4.21]	3.06 [3.06, 3.08] (ov 0.5°)	3.58 [3.54, 3.60]	14% [13, 15]	2.1
90°	5.90 [5.80, 6.00]	5.74 [5.73, 5.76] (ov 3.5°)	5.13 [5.10, 5.18]	13% [11, 14]	3.6
150°	6.86 [6.78, 6.95]	7.04 [7.03, 7.04] (ov 5.2°)	6.18 [6.16, 6.22]	10% [9, 11]	4.7

Delay and update-rate sensitivity. We sweep the optics period (0.15–6.6 s) and latency (0.3 or 3 s) for the LQR and time-optimal controllers (N=150 each; `results/a2_control.json`, 'optics_sensitivity'). With 0.3 s latency, periods up to 2 s still give ≥95% success within 60 s, but the measured 6.6 s period gives only about 40%, because the gyro cannot carry the attitude between sparse, intermittently visible fixes. A 3 s latency reduces success to roughly 30–95%, depending on the period. The latency rows are *not* monotonic in the update rate (fast updates do worse than 2 s updates at 3 s latency), because the correction of fixes still in flight is approximate. They should therefore be read as a warning about long latency, not as design data. It should also be noted that the report's 0.5 Hz figure is not what was measured (6.4 s between fixes).

5.4 The Thresholds the Report Left Undetermined

Table 4 computes these thresholds, and Figure 7 compares them. For imaging, the limit is $\omega \le 0.5\,\mathrm{IFOV}/t_e$, which gives 6.2 mrad/s at 5 ms (1.9 mrad/s at 1/60 s). The star tracker needs a blur of $\lesssim 3$ px of 3.9×10^{-4} rad, i.e. 0.058 rad/s at 20 ms (3.9 mrad/s at 1 px, 100 ms). The sun sensor must keep the pad inside its 55° field between updates, which allows 0.38 rad/s at 2 Hz but only 0.043 rad/s at the measured 6.6 s update period. The report's 0.001 rad/s (A07) is therefore not a consistent bound.

Table 4: Spin-rate thresholds the report left undetermined (computed from IFOV, pixel scale, exposure and update period).

Quantity	Limit [rad/s]	Basis
S0→S1: sun-sensor acquisition	0.3000	sweep rate with smear-aware sigma; FOV limit 0.38 at 2 Hz updates, 0.04 at the measured 6.6 s
S1→S2: star tracker usable	0.0580	3 px blur at 20 ms exposure (0.0039 at 100 ms, 1 px)
S2→S3: imaging	0.0062	0.5 px smear at 5 ms (0.0019 at 1/60 s)
report derivation	0.0010	0.0001 rad / 0.1 s (inconsistent, A07)

6 Estimation

6.1 The Report's Filter Under a Consistency Test

The report's two-state EKF $[\theta, \omega]$ uses a gyro plus $B\cos\theta$ and $B\cos(\theta+\pi/2)$ magnetometers every 2 s, with noise values described as 'arbitrary' (gyro 1 dps, 5 and 6 nT); it was re-implemented here. The normalised estimation error squared

$$\epsilon_k = (\hat{x} - x)^\top P^{-1} (\hat{x} - x), \tag{4}$$

averaged over N independent runs, must lie in $[\chi^2_{Nn}(0.025), \chi^2_{Nn}(0.975)]/N$ [16], where n is the state dimension. For $n = 2$ and $N = 200$ the interval is [1.83, 2.18]. Even when the truth equals the report's

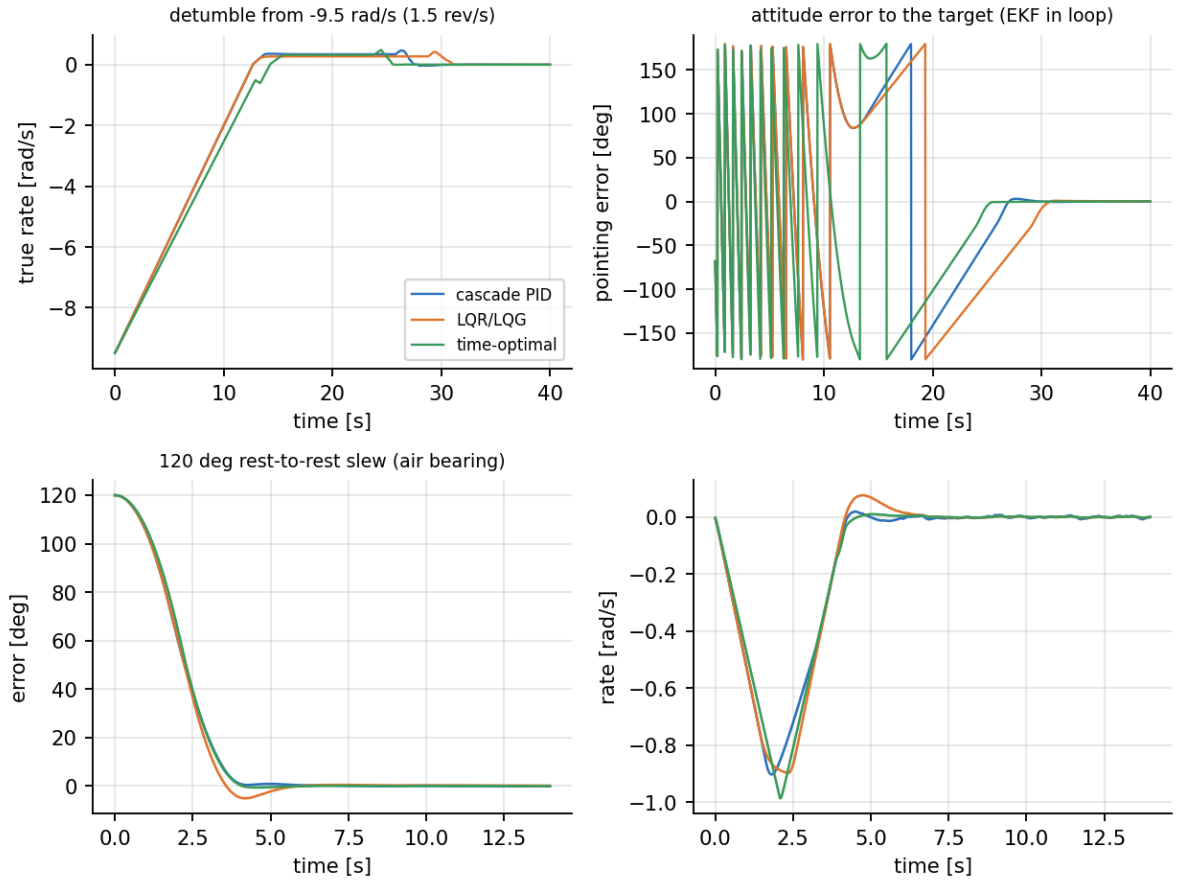


Figure 5: Example trajectories with the same initial conditions and the EKF in the loop. Top: detumble from -1.5 rev/s. Bottom: 120° rest-to-rest slew on an air-bearing plant.

own assumptions, the mean NEES is 0.32 (pessimistic by $\approx 6\times$). With ICM-20948 gyro noise and a 2-LSB (0.3 μT) magnetometer it is 187 (grossly over-confident), and 188 with a 3 dps bias. Its 5 nT magnetometer noise implies 0.010° of heading noise, which is 0.30° below one sensor LSB. The statement that "errors stay within the sigma bounds" is not a consistency test.

6.2 Redesigned Filter

The redesigned filter uses the state $x = [\theta, b]^\top$ and is driven by the gyro:

$$\hat{\theta}^+ = \hat{\theta} + (g - \hat{b})\Delta t, \qquad F = \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix}, \qquad Q = \mathrm{diag}(n_d^2 \Delta t, \sigma_b^2 \Delta t), \tag{5}$$

where g is the gyro reading and n_d is the gyro angle-random-walk density. Sun-sensor and star-tracker fixes z (period 0.5 s, latency τ) are applied by the *measurement-advance* method, with innovation $\nu = z - \hat{\theta}(t - \tau)$ and $S = P_{11} + R + n_d^2\tau + \tau^2 P_{22}$. The delay is thus handled with the stored estimate at the measurement epoch, and the extrapolation uncertainty enters R (conservative, and positive-definite by construction; cf. [14, 15]). Fixes are gated at 6σ, the filter is re-initialised after four consecutive rejections, and R includes the motion smear $\sqrt{(\omega t_{int})^2/12}$.

Two implementation errors found by the consistency tests are worth recording. (a) A first version that applied an extrapolated-gain covariance decrement lost positive-definiteness, and was replaced by the form above. (b) Fixes still in flight were compared with stored epochs that did not yet contain the corrections already applied, which double-counted information; before the fix, a 3 s latency gave 6% mission success and a mean NEES of order 10^4. The corrections are now carried into the stored epochs, and on (re-)initialisation the stored epochs are re-based on the new fix using the integrated gyro.

The filter was evaluated by Monte Carlo ($N = 500$, seed-fixed, with a 0.3 rad/s sweep so that fixes arrive intermittently). The mean NEES is 2.13 against [1.83, 2.18] (77% of time steps inside the interval), and the

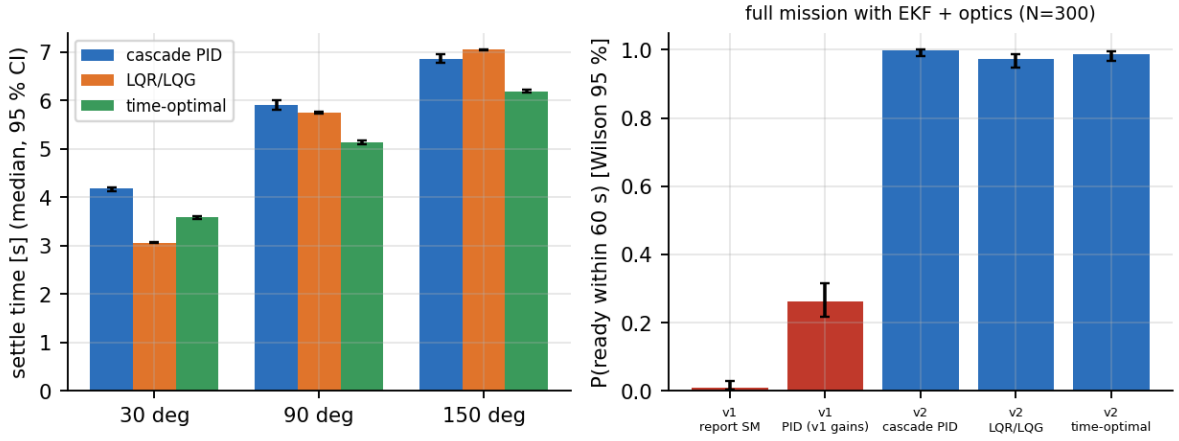


Figure 6: Slew and mission statistics for the controllers. Left: slew settle times (median, 95% CI). Right: probability of being imaging-ready within 60 s for the five hardware/controller combinations of Table 2.

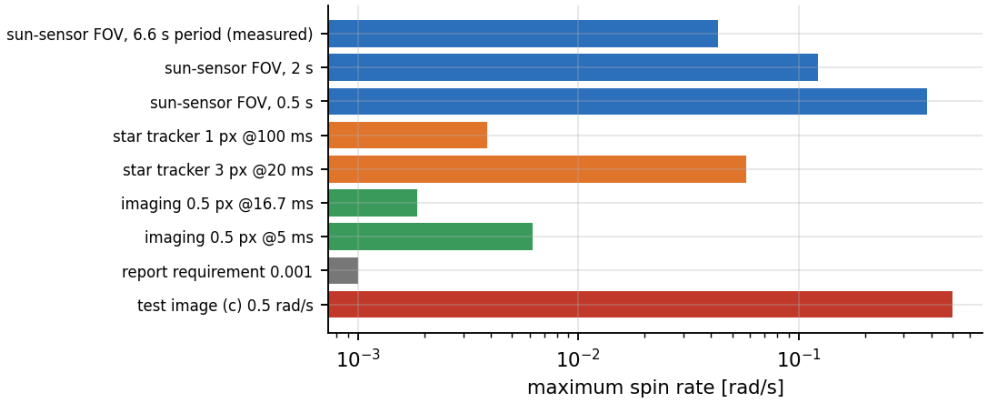


Figure 7: Maximum spin rates implied by each sensor and by imaging, against the report's requirement and its test image (c).

mean NIS is 1.004 against [0.983, 1.017] over 26561 updates. The test also has power: $R \times 0.1$ gives an NIS of 6.8, $R \times 10$ gives 0.13, and $Q \times 0.01$ gives 1.68 (Fig. 8).

6.3 Ablations and Joint Calibration

Figure 9 shows the estimator ablation. The median RMS attitude error over a 90 s sweep is 0.102 [0.098, 0.110] deg for the full filter. Without a bias state it is 0.76 [0.71, 0.91], even with the report's rest-calibrated gyro (7× worse), and 25.8 [21.4, 30.1] with the datasheet zero-rate offset. It is 5.5 [5.5, 5.5] when the 0.3 s latency is ignored, 0.100 [0.095, 0.106] with the star tracker only, 1.11 [1.00, 1.18] with the sun sensor only, and 5 [4, 6] for the as-built optics. Pure gyro integration is useless within the sweep (100 [92, 104]).

On the real Fig. 37 samples (n =21 paired points before the star-tracker misidentification; Fig. 10), the raw ST–SS discrepancy is 5.0° RMS. A gain/offset calibration of the sun sensor against the star tracker reduces it to 1.5° in leave-one-out cross-validation (95% bootstrap [0.9, 1.9]). This is a joint use of the heat pad and the star field that needs no new sensors.

7 Star Identification on the Real Star Field

Map. The 56 star pixel coordinates in the team repository were converted to (ϕ, θ) using the repository's own constants (a 102°×67° reference image, 4608×2592 px). The field spans $-65° \ldots 7°$ in ϕ but only 10° in θ (Fig. 11). The committed `star_table.npy` (46 rows) cannot be reconciled with the CSV (56 rows, 6 removed → 50), so the documented construction was used.

Degeneracy. At the report's tolerance of 0.004 rad, 99% of all star-pair distances have a coincident partner,

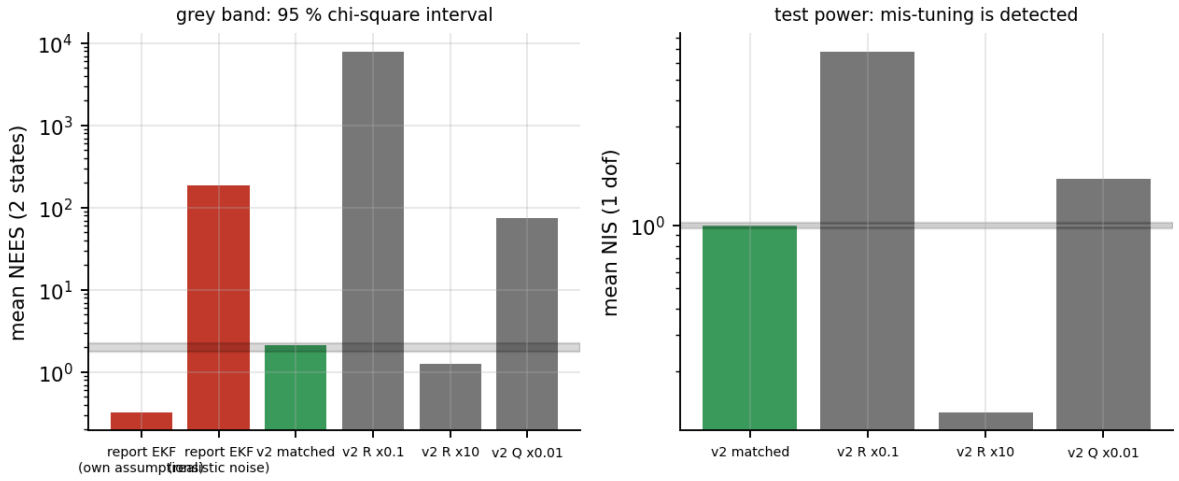


Figure 8: Mean NEES and NIS for the report's and the redesigned filters. The report's filter fails even under its own assumptions; the redesigned filter lies near the bounds, and deliberate mis-tuning is detected.

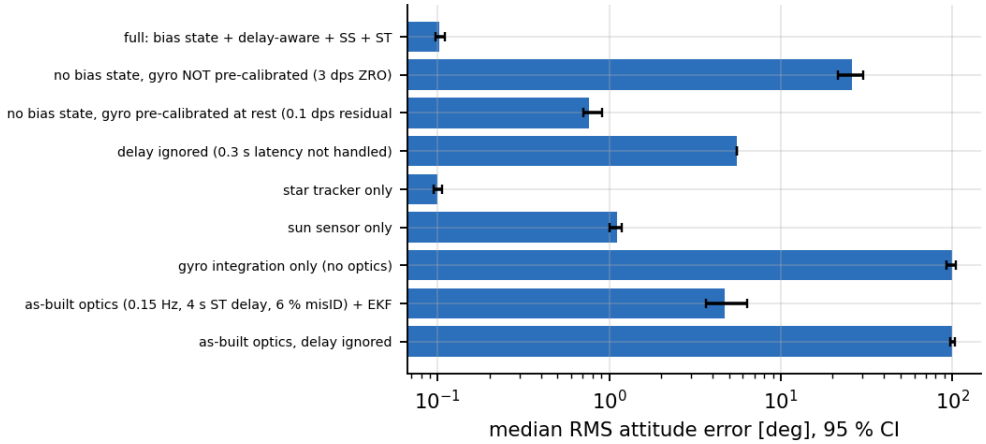


Figure 9: Estimator ablation: median RMS attitude error with 95% CI (log axis) for each filter variant.

and 92% of stars share their first-neighbour distance with another star (3.4 extra candidates on average; median nearest-neighbour distance 0.028 rad). The unique match therefore rests on the later '≥ 5 distance matches' check, which behaves correctly on complete, noise-free data (100% self-match).

Simulation. Images are rendered from the real map with 0.5 px centroid noise, a catalogue error of 0.002 rad, 90% detection and 0.5 false stars per image. The report's matcher (a line-by-line port of `signed_match_to_lookup`) returns the wrong star in 4.9% [3.6, 6.6] of images with the code's 30° crop and 5.1% [3.7, 6.8] with the report's 20° crop (no result 7.6% [6.0, 9.7]), and is correct in 93.6% [91.7, 95.1] and 87.3% [84.8, 89.5] respectively. In pose voting, each (observed, catalogue) pair votes for an offset, and the result is accepted if the peak has ≥ 4 votes and beats every peak at a different ϕ by 2. Pose voting gives 100.0% [99.5, 100.0] and 99.9% [99.3, 100.0] correct with 0.0% [0.0, 0.5] wrong (Table 5, Fig. 12). It tolerates 60% detection (100% against 64%) and a 0.004 rad catalogue error (88% correct and never a wrong star, against 48% correct and 22% wrong for the report). It exploits the one-axis constraint (a pure translation), so it is a special case of geometric voting [10]. As for *speed*, in this port the matcher core takes ≈0.56 ms and pose voting 2.80 ms per image, so the report's 3–5 s is not spent in the matching. Additionally, a triangle area is a function of its three side lengths, so 'area plus distance' adds an index, not information [11].

Field design (dropped). A simulated-annealing design of the 56 positions under coverage and spacing constraints lowers the difference-vector ambiguity cost from 546 (random) and 783 (real field) to 90. However, the identification yield under stress (12° crop, 70% detection, 1 false star/image, 0.003 rad map error) is 62.7% [58.8, 66.6] for random fields with the same spacing (mean of eight), 54.4% [49.4, 59.3] for the real field and 64.6% [61.4, 68.4] for the optimised fields (mean of eight). The annealed layouts beat the *real* field by roughly ten points, but random fields with the same spacing already reach about the same yield, so the gain comes from uniform coverage (stars per window) and not from the uniqueness criterion. Once pose voting is used,

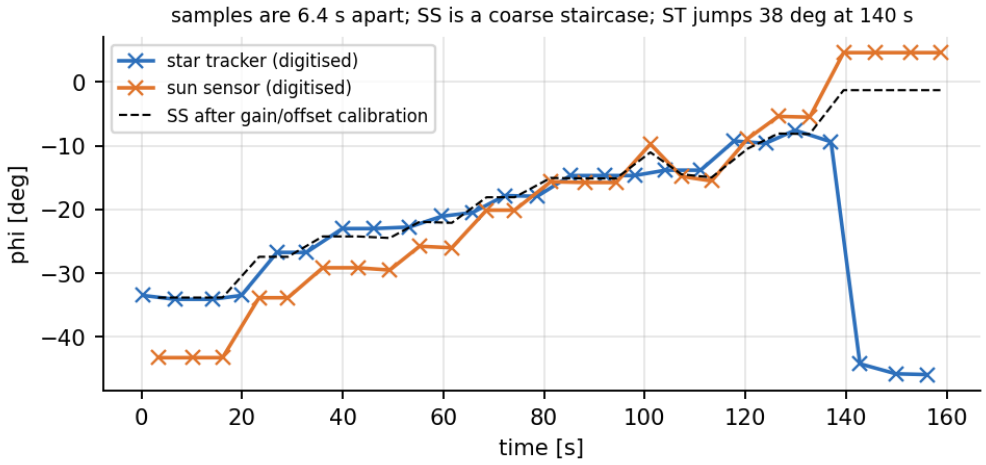


Figure 10: Re-analysis of the report's Fig. 37: samples every 6.4 s, the staircase sun sensor, the 38° star-tracker jump, and the gain/offset-calibrated sun sensor.

the yield is limited by the stars per window and the detection probability. The uniqueness-design idea is therefore dropped, and the practical advice is to space the stars evenly and add more of them where the real field has gaps.

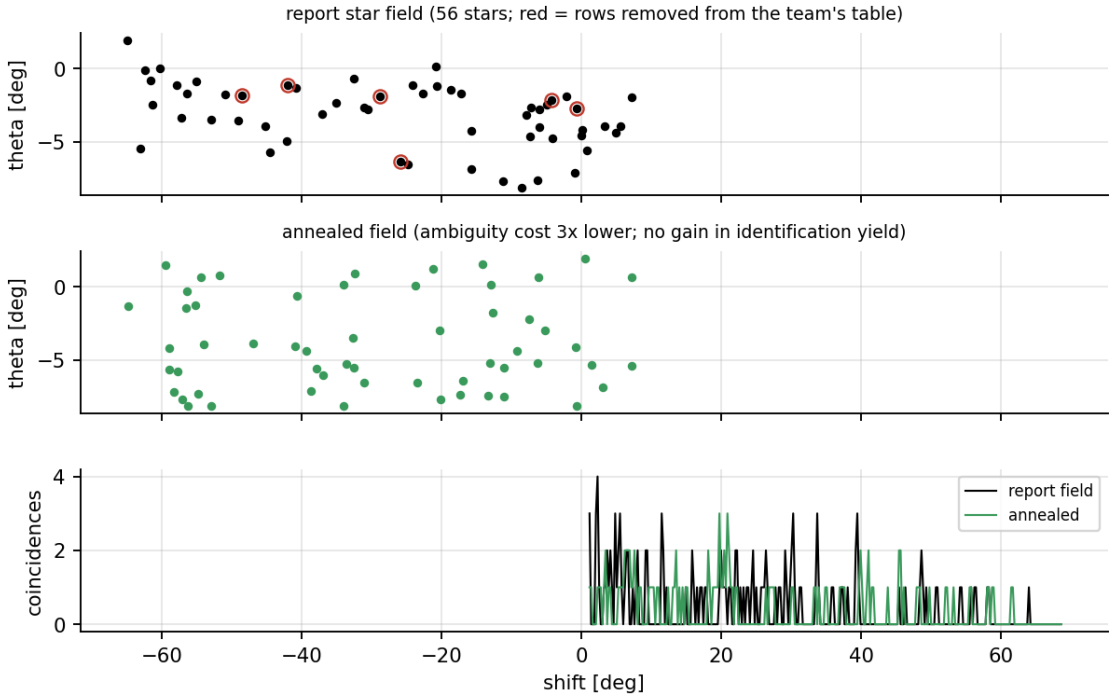


Figure 11: Star-field layouts and their ambiguity. Top: real field. Middle: annealed field. Bottom: shift-coincidence counts at 0.004 rad tolerance.

8 Magnetic Subsystem and Test Coil

Rods. For a rod of diameter d and length l with N turns of 30 AWG (0.338 Ω/m), the dipole is

$$m = \mu_{eff} N I A, \qquad \mu_{eff} = \frac{\mu_r}{1 + N_d(\mu_r - 1)}, \tag{6}$$

where A is the rod cross-section, N_d is the prolate-spheroid demagnetising factor, and $\mu_r = 200$ is assumed. The 493-turn rod is 3.4 Ω and draws 1.47 A, giving $m \approx 1.14$ A m^2 (the report measured 0.95 by a position-

Table 5: Star identification on the report's real star map (simulated images: 0.5 px centroid noise, 0.002 rad catalogue error, 90% detection, 0.5 false stars/image). Wilson 95% intervals.

Crop	Algorithm	correct	wrong star	no result	ms/image
30deg_crop(code)	report	93.6% [91.7, 95.1]	4.9% [3.6, 6.6]	1.5% [0.9, 2.6]	0.56
30deg_crop(code)	hough	100.0% [99.5, 100.0]	0.0% [0.0, 0.5]	0.0% [0.0, 0.5]	2.80
30deg_crop(code)	pairvote	100.0% [98.1, 100.0]	0.0% [0.0, 1.9]	0.0% [0.0, 1.9]	8.49
20deg_crop(report text)	report	87.3% [84.8, 89.5]	5.1% [3.7, 6.8]	7.6% [6.0, 9.7]	0.65
20deg_crop(report text)	hough	99.9% [99.3, 100.0]	0.0% [0.0, 0.5]	0.1% [0.0, 0.7]	2.15
20deg_crop(report text)	pairvote	99.5% [97.2, 99.9]	0.0% [0.0, 1.9]	0.5% [0.1, 2.8]	4.33

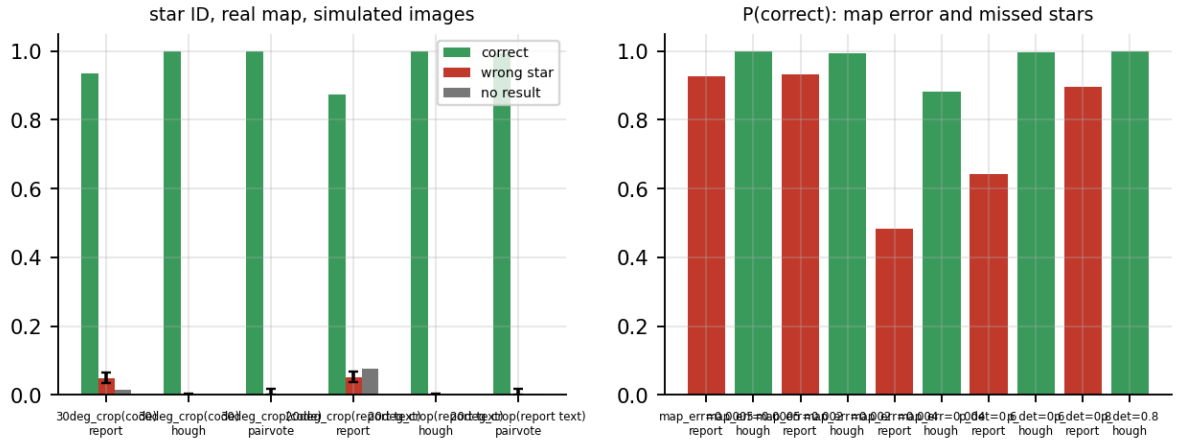


Figure 12: Star-ID outcome rates on the real map, and their sensitivity to catalogue error and missed stars.

fitting method). A 1500-turn, 80 mm redesign gives 1.02 A m^2 at 0.43 A and 2.2 W. Since m scales with μ_{eff} and μ_r is unverified, the moment must be measured.

Drive electronics. A bipolar H-bridge (MOSFET or integrated driver) with current sense, hardware PWM at $\geq$ 20 kHz and a flyback/TVS path replaces the relays. At 2 W per torquer and 0.4 A for the 1500-turn rod this is a simple circuit, whereas the old relay design could not give proportional or reversed dipoles.

Test coil. A Biot–Savart summation over finite straight segments for a square pair gives an optimum separation of $0.546\,a$ (literature 0.5445), where a is the side length. For the built 12 in coils at 14 in, the field varies by 26% p-p over the 10×10×15 cm body. A 0.6 m pair at 0.327 m gives 0.6% and needs 55 A-turn for 150 μT (e.g. 25 turns of AWG 18 at 2.2 A, 5.6 V, 12 W). The report's 'N cancels' argument holds only for a voltage source.

Torque authority and de-saturation. With the identified friction (4.1×10^{-4} N m), the available magnetic torque (mB, with 0.95–1.6 A m^2 at 90 μT) is only 0.21–0.35 of the friction. The testbed therefore cannot show magnetic detumbling at all, whereas the same torquers exceed an air-bearing residual by 14×. Meeting Requirement A by magnetic torque alone in 60 s would need 2.1 A m^2 at 90 μT.

Momentum management (S4). On an air bearing (10^{-5} N m friction, with the field at 40–100% of nominal and a 10° direction error to represent Table 10's gradient and misalignment), a wheel at $h = 0.02$ N ms must be dumped. The open-loop concurrent S4 of the report's concept loses pointing in 100% of trials. Closed-loop attitude control plus magnetorquer momentum feedback keeps the pointing within 0.19° rms and reaches 20% of $|h|$ in 325 [307, 335] s (68% within 500 s), close to the bound $H/(mB) \approx 317$ s. A coordinated LQ design reaches the same time (326 [307, 335] s) with a worse attitude transient (6.3°). Coordination therefore gives no dump-time benefit, because the limit is mB; it is dropped as an improvement (§10). Figure 13 summarises the magnetic results.

9 Imaging

The optics have an IFOV of 62 μrad, a GSD of 0.31 mm at 5 m and an FOV of 14.3° × 10.8°. The 25 cm sphere spans 806 px and a 1 mm feature is 3.2 px. The diffraction limit is an Airy radius of 1.04 px at f/2.4, and the total depth of field is 0.59 m at 5 m (2-px circle of confusion). A 0.7 mm resolved feature is 2.3 px,

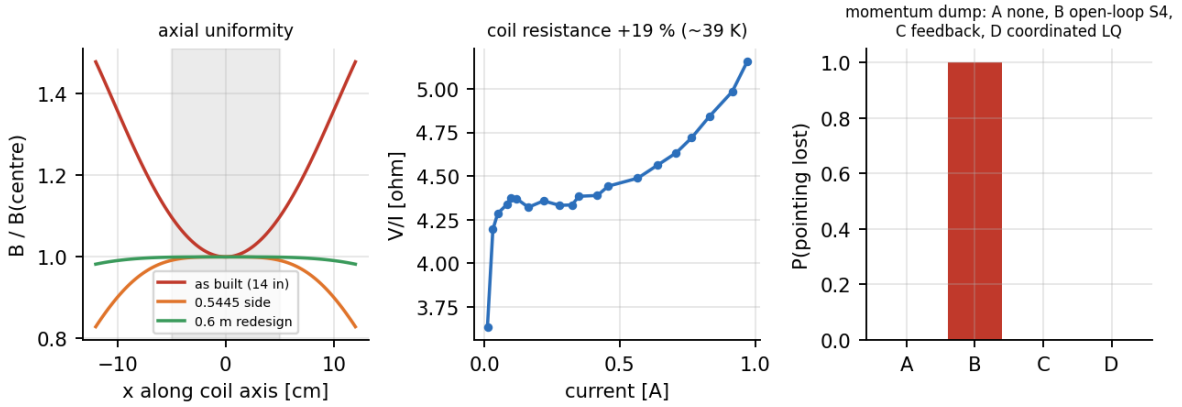


Figure 13: Magnetic subsystem results. Left: axial coil uniformity (shaded: body extent). Middle: measured coil resistance rise (Table 10 of the report). Right: probability of losing pointing during the momentum dump.

so the report's static resolution claim is consistent. The 60 fps video, however, needs a binned mode on this sensor (unverified), in which 1 mm is 1.6 px, below Nyquist.

The smear MTF is $\mathrm{sinc}(fs)$ at spatial frequency f. At the 1 mm line-pair frequency, MTF$= 0.5$ at $s = 3.9$ px, so the 0.5 px budget is conservative. To test this, we render a crater scene (1 mm craters, Poisson noise plus 3 e^- read noise), apply exact uniform motion blur, and compare (i) a 10 ms exposure, (ii) an exposure shortened to 0.5 px of smear, (iii) K gyro-registered short frames (registration error 0.15 px, growing with $\sqrt{1 + k/10}$, with K limited by the dwell time in the 14.3° field), and (iv) a strobe 20× brighter. The fidelity ρ (correlation with the sharp scene) and the crater detection at 1% false alarm give the maximum rate with $\rho \geq 0.8$ and $P_d \geq 0.9$. At 2×10^5 e^-/px/s, the maximum tested rate meeting the criterion is none for the 10 ms exposure (rad/s, 'none' = never), 0.05 for stacking and 0.02 for strobing; at 8×10^5 it is 0.02 (naive) and 0.2 (stacked) (Fig. 14). Stacking and strobing therefore matter only if the rate cannot be held below ~6 mrad/s, which the hold controller achieves.

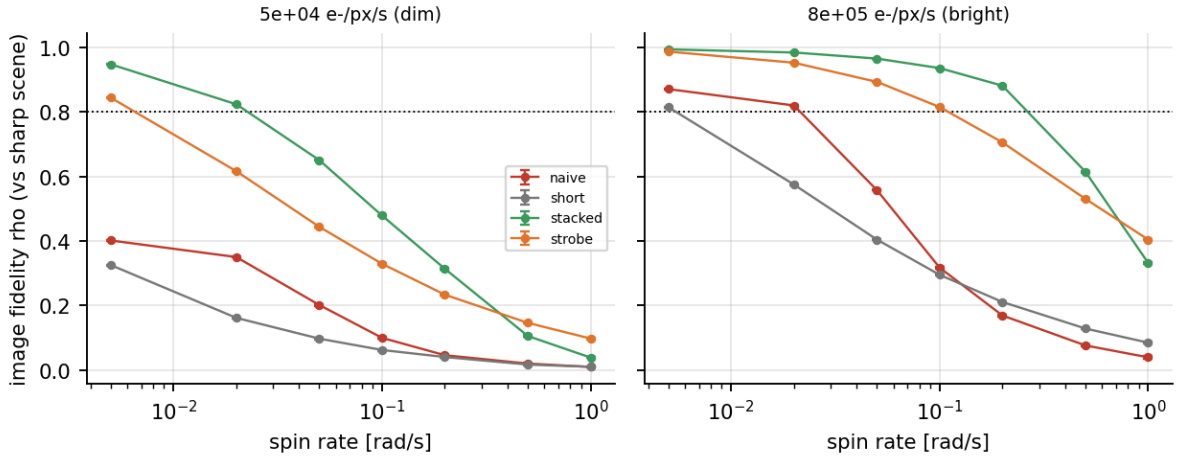


Figure 14: Image fidelity versus spin rate for four imaging strategies at two illumination levels (95% CI over 60 scenes).

10 Novel-Methods Workstream

Candidate improvements were evaluated against the redesigned baseline by deterministic Monte Carlo with confidence intervals, and screened against the literature (Table 6). The verdicts use three terms: HOLDS-REPLICATED (the gain is real, but the idea is already published), HOLDS-APPLIED (the gain is real and the idea is standard, so the value is in the quantified application), and DROPPED.

N1 Time-optimal manoeuvres. These give faster rest-to-rest slews, 13% [11, 14] at 90° and 10% [9, 11] at 150° against the PID (§5), but no detumble gain, since torque limits every controller. The method is

Table 6: Novel-methods workstream: verdicts against the literature (S-numbers in `research/sources.md`).

ID	Idea	Verdict	Literature / why
N1	time-optimal (bang-bang with torque saturation) detumble and slew	HOLDS-REPLICATED	Time-optimal attitude manoeuvres with wheel limits are established (S25-S29, incl. flight results >10 % shorter slews)
N2	reaction-wheel + magnetorquer momentum management in one optimal scheme	DROPPED (as an improvement)	Optimal wheel/magnetorquer momentum unloading is established (S20-S22: Camillo & Markley 1980, Chen et al. 1999, Chen et al. 2010)
N3	sensor fusion: gyro-bias estimation from optical fixes; delayed/multi-rate updates; joint heat-pad + star-field calibration	HOLDS-REPLICATED	Gyro bias from star-tracker data (S52, S53), delayed/out-of-sequence updates (S59, S60), MLX90640 sun sensing with MEKF (S50, S51), MEKF (S57, S58)
N4	faster / more robust star identification (pose voting exploiting the 1-axis constraint)	HOLDS-APPLIED	Geometric voting (S46), planar-triangle ID (S45), pyramid (S42); a translation Hough vote is a special case
N5	star-field design for uniqueness (difference-vector multiplicity annealing)	DROPPED (uniqueness criterion)	no direct star-field-design-for-uniqueness paper found in the survey (S48-S49 optimise catalogue databases); Golomb/Costas-type ideas exist in other fields
N6	exposure / shutter synchronisation, gyro-registered stacking, strobed illumination	HOLDS-REPLICATED	Gyro-aided deblurring (S61), MEMS-gyro restoration of star images (S62), attitude-correlated frames (S63), rolling-shutter compensation (S64-S66)
N7	testbed: air bearing and coast-down friction identification	HOLDS-APPLIED	Air-bearing simulators (S30-S32), low-cost testbeds (S33-S36), friction identification (S37, S38)

established [7, 8]. **N2 Wheel+MTQ.** Dropped (§8). **N3 Fusion.** The bias state, delay-aware updates and SS–ST calibration give the large gains of §6; all are standard [12, 13, 14]. **N4 Star ID.** Pose voting removes wrong IDs in simulation; the speed claim is dropped. **N5 Field design.** Dropped. **N6 Smear defeat.** Known [17, 18]; the operating envelope is quantified here. **N7 Testbed.** Fitting pulse-time series instead of three stop times shrinks the friction uncertainty from -32/+71% to ±0.7% (single Hall magnet) or ±0.001% (360 cpr) in simulation, with the caveat that real bearings add Stribeck and stick-slip model error. An air bearing (residual 10^{-6}–10^{-5} N m [9]) is required to test the magnetics, and it must be levelled to 0.020° for a 3 mm CoM offset.

11 v2 Baseline and Test Campaign

Architecture. The v2 baseline is a 2U spacecraft (100×100×227 mm) with the same sensor suite, a wheel with an encoder and closed speed loop, H-bridge torquers, a 4S 1300 mAh pack, a ±1000 dps gyro range, a strobe-capable target light, and an air-bearing platform. The 1U option cannot route three cameras and an 80 mm wheel (component volume 553 cm^3; 2U fill 24%). The mass is 858 g (987 g with 15% margin; < 1 kg). The energy for the 21 min campaign is 6.5 Wh against 19.2 Wh (margin 2.4×), and the peak current is ≈2.1 A against the report's 2 A assumption. All loads are assumed and will be replaced by test T9, and the battery energy density of 140 Wh/kg is derived from verified pack data. The requirements, mass and power budgets are given in Tables 7, 8 and 9.

Table 7: Re-baselined requirements (verification and acceptance rules).

ID	Requirement	Verification / acceptance	Repairs
R-DET-1	From a spin of 1.0 rev/s (range 0.5-1.7 rev/s), the ADCS shall reduce the rate below 10 deg/s within 60 s	Test on air bearing (residual <1e-5 N m), ADCS-on vs ADCS-off paired runs; N>=20 runs, >=95 % pass, one-sided 95 % lower Wilson bound >=85 %; ADCS-off runs must fail	A03, A04
R-DET-2	From 10 deg/s, reduce to <= 0.5 deg/s within 60 s	Test on air bearing; as R-DET-1	A03
R-ACQ-1	Acquire attitude (EKF initialised, sigma < 1 deg) within 30 s of reaching 0.3 rad/s from unknown attitude	Test with randomised start angle (>=20 runs); median <= 20 s, p95 <= 30 s	A24
R-PNT-1	Hold the target within +/-3.5 deg (target sphere inside the 10.8 deg short axis) with rate <= 6.2 mrad/s for >= 10 s (stretch: 60 s)	Test; attitude from encoder/table reference; fraction ready >= 0.95 over 3 x 20 runs	A07, A08, A23
R-IMG-1	Image 1 mm features at 5 m: GSD <= 0.33 mm (k=3), smear <= 0.5 px, static MTF50 >= 1 mm line-pair frequency	Slanted-edge MTF at operating exposure and mode; image of calibration chart at 5 m; MTF at 0.155 cyc/px >= 0.4; chart groups to 0.7 mm resolved in 3/3 frames	A19, A08
R-IMG-2	Exposure/illumination shall deliver >= 20 dB SNR at the chosen exposure without saturation	Photometric test with strobe/continuous lamp; SNR measured on grey patch >= 20 dB at <= 5 ms	A07
R-ADS-1	EKF attitude error <= 0.1 deg (1-sigma) during hold, NEES and NIS consistent with chi-square 95 % intervals over >=500 simulated and 5 hardware runs	Analysis + replay of logged data; time-averaged NEES within the interval or conservative; NIS mean within [0.97,1.03]	A21
R-ADS-2	Gyro range >= 1000 dps (+/-17.5 rad/s) with saturation flag logged; bias estimated online	Inspection + test with 1.7 rev/s spin; no saturation flag in 20 runs; bias estimate converges within 10 s	A04
R-STR-1	Star identification: P(correct) >= 0.99, P(wrong) <= 0.5 % per attempt over >=1000 stored images; processing <= 1 s on the Pi	Test on stored images + analysis; Wilson upper bound on P(wrong) <= 1 %	A22
R-SUN-1	Sun sensor update >= 1 Hz (full frame at refresh >= 2x), residual vs encoder <= 0.5 deg after gain/offset calibration	Rotary table with encoder; RMS <= 0.5 deg over 3 sweeps	A20
R-DMP-1	Reduce wheel momentum by 0.02 N m s (10.7 rad/s-equivalent) within 400 s while holding attitude within 1 deg	Test on air bearing, B >= 90 uT, m >= 1.0 A m^2; N>=10 runs, >=90 % reach 20 % of initial |h| within 400 s	A13
R-MTQ-1	Torquer pair: dipole >= 1.0 A m^2 each at <= 6 W, bipolar drive via H-bridge, flyback protected, thermal cut-off at 80 C	Helmholtz-cage moment measurement with uncertainty budget; m within +/-15 % of design in 3 runs	A09, A10, A11
R-WHL-1	Wheel: torque >= 1 mN m, momentum capacity >= 0.04 N m s, speed measured to 0.1 RPM, command resolution <= 0.2 RPM (16 bit)	Spin-up/coast-down with encoder; resolution test: 0.2 RPM step visible in encoder	A02

ID	Requirement	Verification / acceptance	Repairs
R-PWR-1	Operate >= 15 min with >= 2x energy margin and a logged power profile; peak current <= 4 A; no brown-out of the 5 V rail	Bench + full-system run with logger; min 5 V rail >= 4.75 V in all runs	A18
R-MAS-1	Mass <= 1.0 kg with 15 % margin; envelope 100 x 100 x 227 mm (2U)	Scale + inspection; mass measured <= 1.0 kg	A25
R-TB-1	Testbed residual disturbance torque <= 1e-5 N m (air bearing), levelled to <= 0.02 deg for a 3 mm CoM offset	Coast-down with 360+ cpr encoder, fit Coulomb+viscous; Coulomb torque estimate 95 % CI upper bound <= 1e-5 N m	A03, A16, A17
R-TB-2	Magnetic test field >= 3x Earth (90 uT) uniform to <= 10 % p-p over the 10 x 10 x 15 cm body, reproducible to 10 %	3-axis grid map at 3 currents, 3 days; p-p <= 10 %, day-to-day <= 10 %	A14, A15

Table 8: v2 mass budget (g). Status: DERIVED from CAD/verified data or ASSUMED catalogue value.

Item	Mass	Status
structure (bottom plate PETG + 3 Al plates + brass standoffs, 2U frame)	250	DERIVED from CAD (cad_inertia) scaled to 227 mm
reaction wheel + BLDC motor + encoder + mount	150	ASSUMED
2 magnetorquers (1500 turn, 80 mm rod)	68	DERIVED (a5)
PDB with H-bridge drivers	45	ASSUMED
Raspberry Pi 5	46	ASSUMED (catalogue)
Arduino-class MCU	10	ASSUMED
IMU (ICM-20948) x2 + breakout	8	ASSUMED
HQ camera + 25 mm lens	62	ASSUMED
star camera (Module 3 wide)	8	ASSUMED
MLX90640 sun sensor	4	ASSUMED
battery 4S 1300 mAh	137	DERIVED from verified energy density
harness, fasteners, brackets	70	ASSUMED
total	858	+15% margin = 987

Table 9: v2 power budget by mode (ASSUMED loads, 87% converter efficiency; replaced by measurement in test T9).

Mode	load [W]	battery [W]	current at 14.8 V [A]
detumble (wheel high torque)	26.9	30.9	2.09
capture / slew	16.9	19.4	1.31
imaging hold	10.9	12.5	0.84
S4 dump	15.3	17.6	1.19

Test campaign. Table 10 lists ten tests with their sample sizes and acceptance rules. Every pass/fail rule uses a Wilson lower bound or a CI, and every dynamic test pairs ADCS-on with ADCS-off runs. The schedule (Table 11) is 14 weeks with a margin week.

12 Discussion

12.1 Lessons Learned

The audit and redesign point to a small number of practices that would have changed the original outcome:

1. **Close the units loop.** Two critical errors (inertia ×2, RPM ×10) were found by recomputing every number; one unit check would have changed the actuator conclusion.
2. **A requirement must be able to fail.** Requirement A passed with the ADCS off. Every dynamic test needs an ADCS-off control run and a bearing whose disturbance is $\ll$ the control torque.

Table 10: v2 test campaign.

ID	Test	Objective	N	Acceptance
T1	Static optical characterisation	GSD, MTF, focus, mode, exposure (R-IMG-1/2)	3 sessions	slanted-edge MTF, 5 m chart
T2	Gyro/IMU calibration	six-position, Allan variance, scale, range (R-ADS-2)	12 h static + rotary table	bias instability and ARW published
T3	Sun sensor and star tracker calibration	angle vs rotary-table encoder (R-SUN-1, R-STR-1)	>=1000 images	RMS and P(wrong)
T4	Friction/air-bearing characterisation	coast-down fit (R-TB-1)	5 runs x 3 speeds	Tc upper bound <= 1e-5 N m
T5	Wheel characterisation	torque, resolution, momentum, coast (R-WHL-1)	10 runs	resolution, time constants
T6	Magnetorquer and Helmholtz mapping	dipole, field uniformity (R-MTQ-1, R-TB-2)	3 days	m within 15 %, p-p <= 10 %
T7	Detumble campaign	R-DET-1/2 paired ADCS on/off	>=20 runs each	Wilson bounds
T8	Acquire, point, image	R-ACQ-1, R-PNT-1, R-IMG-1	>=60 runs	ready fraction; image chart groups
T9	Power and endurance	R-PWR-1	3 x 15 min	rail min voltage
T10	Momentum dump	R-DMP-1	>=10 runs	time to 20 % of |h|

Table 11: Fourteen-week schedule.

Weeks	Requirements	Work
W1-2	R-*	Re-baseline requirements, order long-lead items (encoder wheel, air bearing, H-bridge, strobe)
W3-4	R-IMG-1, R-ADS-2, R-SUN-1	Bench: IMU/gyro, optics MTF, sun sensor and star tracker calibration (T1-T3)
W5-6	R-TB-1, R-TB-2	Testbed: air-bearing platform + encoder, friction coast-down (T4); Helmholtz rebuild (T6)
W7-8	R-WHL-1, R-MTQ-1	Wheel and torquer characterisation (T5, T6); PDB v2 with drivers
W9-10	R-ADS-1, R-STR-1	Software: EKF (bias, delay), pose-voting star ID, controller, S4; replay on logged data
W11-12	R-DET-*, R-PWR-1	Integration, power profile and endurance (T9), detumble campaign (T7)
W13	R-ACQ-1, R-PNT-1, R-DMP-1	Acquire/point/image campaign (T8), momentum dump (T10)
W14	-	Margin week, report

3. **Do not judge estimator quality by eye.** 'Inside the sigma bounds' accepts both a filter 90× overconfident and one 6× under-confident; NEES/NIS with chi-square intervals and deliberately mis-tuned controls should be used instead.
4. **Know the sensor limits before the test.** A saturated gyro, 6.4 s optical sampling and a coarse sun-sensor staircase were all visible in the data; range, rate and quantisation should be measured first.
5. **Log what will need to be proven.** Exposure, mode, wheel speed, voltage and the gyro full-scale flag were missing, and each cost an inference.
6. **Design for the physics bound.** The de-saturation requirement needed ~ 9 A m^2; momentum dumping is bounded by $H/(mB)$, and no controller changes that.
7. **Screen novelty honestly.** Nearly every candidate already existed; the value is in the quantified, reproducible application and in the negative results.
8. **Treat third-party code as data.** The repository's lookup table did not match its documented map; reading it, rather than running it, was enough to show this.

12.2 Sources of Error and Model Limitations

Several simplifying assumptions in this analysis warrant discussion regarding their effect on the results.

(i) Digitised data. The Fig. 37/38 values are digitised (resolution ≈0.05 rad/s, 0.03 s), and the Hall trace is sparse. The identified friction and wheel parameters carry this resolution in their uncertainty bands.

(ii) Phenomenological wheel model. The wheel/throttle model is phenomenological and identifies only the acceleration limit; I_w/I_t separately is not identifiable.

(iii) Assumed parameters. The component masses, the powers (Pi 5 ≈7 W), the rod μ_r and the gyro bias random walk are assumed.

(iv) One-axis, idealised simulation. The simulations are one-axis and idealised: they include no cross-axis coupling, no rolling-shutter skew (readout time unverified), no COM unbalance on the bearing, and no Stribeck friction.

(v) Estimator consistency. EKF consistency is shown only for simulated Gaussian noise that matches the filter's model. The delay is handled by a conservative approximation, and real sensor noise (non-Gaussian, temperature-dependent) is not tested.

(vi) Simulated star images. The star-ID campaign uses simulated images and a catalogue-error model; the real lens distortion of the 102° lens is not modelled.

(vii) Rod sizing source. The Amin–Lightsey sizing equations [6] could not be read (access denied), so the rod model is first-principles only.

(viii) Costs. The costs of air bearings and strobes are not priced (unverified).

13 Conclusion

This work re-examined the SPACE 584 Fall 2025 one-axis ADCS testbed by auditing the final report, identifying a dynamic model from the report's own digitised data, and redesigning the controller, estimator, star identification, magnetic subsystem and imaging strategy, each evaluated by deterministic Monte Carlo.

At the component level, the results show that several of the report's conclusions follow from arithmetic and unit errors rather than from the hardware: the inertia and wheel-resolution requirement were mis-sized, the gyro was saturated in the headline test, and the report's filter is not consistent even under its own assumptions. The identified model shows that the testbed's bearing friction dominates both the detumble requirement and the available magnetic torque, and that the wheel channel was acceleration-limited.

At the system level, a closed-loop rate/attitude controller with a consistent, delay-aware estimator raises the simulated probability of reaching imaging within 60 s from that of the report's state machine to near certainty on v2 hardware, while a time-optimal controller shortens rest-to-rest slews. Most candidate improvements are replications of published methods, and two do not hold.

In general, the results demonstrate that the testbed, as built, could not have shown the behaviour its requirements asked for, and that the limiting factors were requirement design, units and sensor limits rather than control theory. Future work should build and run the v2 test campaign, starting with an air-bearing platform and an ADCS-off control run for every dynamic test, and should replace the assumed loads, rod permeability and sensor noise with measurements.

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A Reproducibility

`run_all.py` regenerates every number, figure and table from the PDF and the data, with seeds derived from one master seed through named streams. `tests/test_all.py` checks the closed forms, the key results, bit-for-bit determinism and the power of the NIS test. The quick mode runs the same code with a smaller N.

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